



Exploiting Artificial Intelligence in Software Development

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Motivation & Problem Statement



1



Massive Log Volume

Modern systems generate billions of log lines daily

2



Manual Inspection

Infeasible at scale — humans can't keep up

3



Anomaly Detection

Critical for reliability, but existing tools lack interpretability

4



The Goal

Detect anomalies AND explain them in human-readable form

Thesis Goal



Build a general-purpose, user-friendly agentic AI framework

for log anomaly detection — with classification AND natural language explanation



Generalizability

KEY PRINCIPLE

Works across different log formats and system types without retraining



Minimal Fine-Tuning

KEY PRINCIPLE

No (or near-zero) fine-tuning required — prompt-driven by design



Robustness

KEY PRINCIPLE

Resilient to log format changes and distribution shifts

Literature Review

5 KEY PAPERS SURVEYED

		DETECTION SCOPE	APPROACH TYPES
1	LLM-LADE Classification + annotated explanations	Explanation and Detection	LLM-based Framework
2	LogLLM LLM-based log parsing & detection	Detection	LLM-based Framework
3	LogRESP-Agent Agent-based log reasoning pipeline	Explanation and Detection	Agentic
4	Prompt Pattern Catalog Prompt engineering patterns for ChatGPT	—	Prompting
5	LogGPT GPT-based anomaly detection in system logs	Explanation and Detection	LLM-based Framework

KEY FINDING



Most works focus on classification only

Explanation of anomaly root causes remains largely unexplored — this is the gap we try to address.

APPROACH TYPES

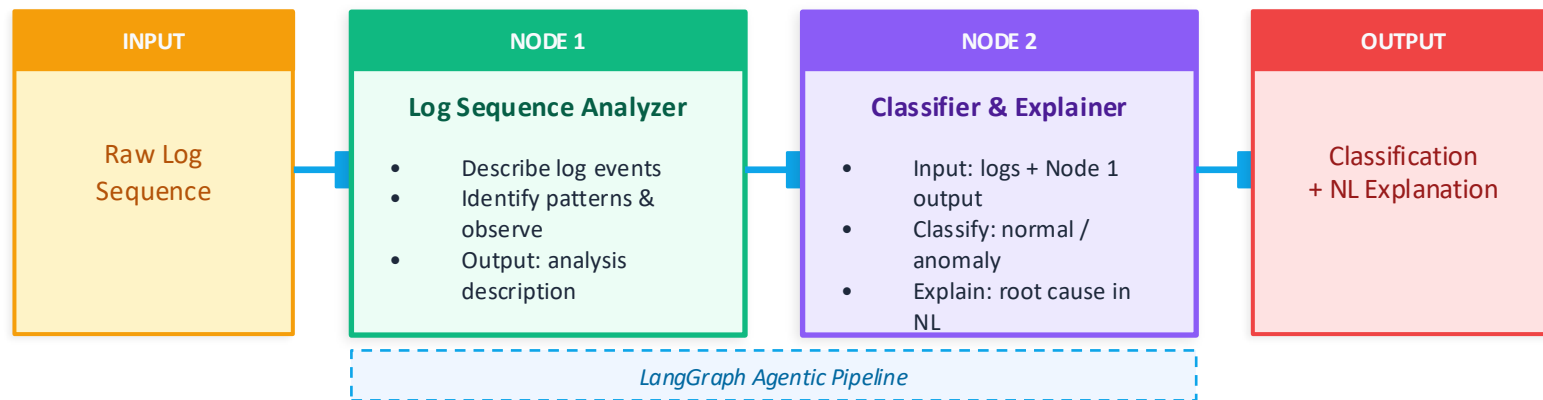
- LLM-based Framework
- Agentic
- Prompting

DETECTION SCOPE

- Detection
- Explanation and Detection

System Architecture

First Prototype — LangGraph Agentic Pipeline



Python

LangChain

LangGraph

Model-Agnostic

MCP / Tools (planned)

Datasets

LogHub Benchmark · 3 datasets · 30 annotated sequences each (LLM-LADE)

HDFS

Hadoop Distributed File System

SOURCE

Distributed storage nodes

SIZE

11 M lines

ANOMALY RATE

~2.9%

EXAMPLE LOGS

```
Receiving block <*> src: <*> dest: <*>
```

```
BLOCK* NameSystem.allocateBlock: <*> <*>
```

BGL

Blue Gene/L Supercomputer

SOURCE

BlueGene/L system logs

SIZE

4.7 M lines

ANOMALY RATE

~7.4%

EXAMPLE LOGS

```
instruction cache parity error corrected
```

```
ddr errors detected and corrected
```

Thunderbird

Sandia National Labs Cluster

SOURCE

Large-scale HPC cluster

SIZE

211 M lines

ANOMALY RATE

~0.7%

EXAMPLE LOGS

```
error detected on node <*>
```

```
kernel: <*>: eth0: transmit timed out
```

Benchmarking (Best)

First Prototype Evaluation · Thunderbird Dataset

① Classification Performance

94.44%

F1 Score

100%

Recall

89.47%

Precision

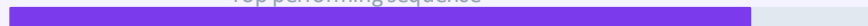
93.33%

Accuracy

② Explanation Quality (text-embedding-3-large)

0.8579 Best Similarity

Top performing sequence



0.7774 Average Similarity

Embedder cosine score



0.6971 Worst Similarity

Lowest performing sequence



Benchmarking (Worst)

First Prototype Evaluation · HDFS Dataset

① Classification Performance

63.64%

F1 Score

50%

Recall

87.50%

Precision

73.33%

Accuracy

② Explanation Quality (text-embedding-3-large)

0.8364

Best Similarity

Top performing sequence

0.7293

Average Similarity

Embedder cosine score

0.6427

Worst Similarity

Lowest performing sequence

Data Augmentation

08



Problem: only 30 annotated sequences per dataset — insufficient for robust evaluation

Approach 1 · In progress

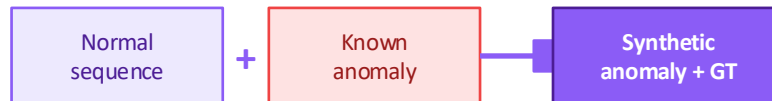
LLM Few-Shot Synthetic Generation



- Script implemented
- First synthetic data already generated
- Uses 3–5 real examples as prompt context
- Generates varied log sequences with labels & explanations
- Need validation strategy to verify generated datasets (log sequence correctness, classification accuracy, and cause explanation quality)

Approach 2 · Planned

Anomaly Injection into Normal Sequences



- Inject known abnormal messages into normal logs
- Creates synthetic anomalies with verifiable root causes
- Ground truth labels are deterministic — no annotation needed
- Enables scalable, controllable evaluation

Next Steps

Planned work for thesis completion

09



01

Validate & Expand Dataset

Review augmented data quality · scale to all 3 datasets

02

Tool Use Support

Extend prototype with MCP / tool calling capabilities

03

Integrate LogLLM

Add full LLM-based log detection framework as an agentic tool

04

Design Pipeline Tools

Build dedicated tools for log parsing, pattern search & retrieval

05

Re-evaluate System

Run full benchmark on improved pipeline with augmented data

Summary

Exploiting Artificial Intelligence in Software Development

COMPLETED

Literature Review

5 key papers surveyed

System Architecture

LangGraph 2-node pipeline

Benchmarking

Thunderbird and HDF5

Data Augmentation

Few-shot synthesis in progress

PLANNED

Expand dataset

Tool use / MCP

Integrate LogLLM

Re-evaluate

Expected contribution: a general-purpose, model-agnostic agentic framework for log anomaly detection that not only classifies but explains root causes in natural language.

Thank You

Questions?

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SEMESTER	Summer 2026